```
from google.colab import files
uploaded = files.upload()
import pandas as pd

df = pd.read_csv('netflix_titles.csv')  # replace with your file name
df.head()

```

Choose Files No file chosen Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to enable.

```
# 1. Basic structure
df.info()

# 2. First few rows
df.head()

# 3. Check column names
df.columns

# 4. Null values
df.isnull().sum()

# 5. Data types of each column
df.dtypes
sum()

```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
/tmp/ipython-input-5-763199930.py in <cell line: 0>()
      1 # 1. Basic structure
----> 2 df.info()
      3
      4 # 2. First few rows
      5 df.head()

NameError: name 'df' is not defined
```

```
import pandas as pd

df = pd.read_csv('netflix_titles.csv')  # Replace with correct file name if needed
df.info()
df.head()

```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 8807 entries, 0 to 8806
Data columns (total 12 columns):
 #   Column        Non-Null Count  Dtype 
---  ------        --------------  ----- 
 0   show_id       8807 non-null   object
 1   type          8807 non-null   object
 2   title         8807 non-null   object
 3   director      6173 non-null   object
 4   cast          7982 non-null   object
 5   country       7976 non-null   object
 6   date_added    8797 non-null   object
 7   release_year  8807 non-null   int64 
 8   rating        8803 non-null   object
 9   duration      8804 non-null   object
 10  listed_in     8807 non-null   object
 11  description   8807 non-null   object
dtypes: int64(1), object(11)
memory usage: 825.8+ KB
```

	show_id	type	title	director	cast	country	date_added	release_year	rating	duration	listed_in	description
0	s1	Movie	Dick Johnson Is Dead	Kirsten Johnson	NaN	United States	September 25, 2021	2020	PG-13	90 min	Documentaries	As her father nears the end of his life, filmm...
1	s2	TV Show	Blood & Water	NaN	Ama Qamata, Khosi Ngema, Gail Mabalane, Thaban...	South Africa	September 24, 2021	2021	TV-MA	2 Seasons	International TV Shows, TV Dramas, TV Mysteries	After crossing paths at a party, a Cape Town t...
2	s3	TV	Ganglands	Julien	Sami Bouajila, Tracy Gotoas,	NaN	September	2021	TV-MA	1 Season	Crime TV Shows, International	To protect his family from a

```
# Remove duplicate rows (if any)
print("Before removing duplicates:", df.shape)
df = df.drop_duplicates()
print("After removing duplicates:", df.shape)

```

```
Before removing duplicates: (8807, 12)
After removing duplicates: (8807, 12)
```

```
# Step 1: Remove Duplicates
print("Before removing duplicates:", df.shape)
df = df.drop_duplicates()
print("After removing duplicates:", df.shape)

# Step 2: Fill Missing Values
df['director'] = df['director'].fillna('Unknown')
df['cast'] = df['cast'].fillna('Unknown')
df['country'] = df['country'].fillna('Unknown')
df['date_added'] = df['date_added'].fillna('Unknown')

# Step 3: Check Again
df.isnull().sum()

```

```
Before removing duplicates: (8807, 10)
After removing duplicates: (8797, 10)
/tmp/ipython-input-14-2299214124.py:7: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-ver
  df['director'] = df['director'].fillna('Unknown')
/tmp/ipython-input-14-2299214124.py:8: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-ver
  df['cast'] = df['cast'].fillna('Unknown')
/tmp/ipython-input-14-2299214124.py:9: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-ver
  df['country'] = df['country'].fillna('Unknown')
/tmp/ipython-input-14-2299214124.py:10: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-ver
  df['date_added'] = df['date_added'].fillna('Unknown')
```

	0
type	0
director	0
cast	0
country	0
date_added	0
release_year	0
rating	4
duration	3
listed_in	0
description	0

dtype: int64

```
df.isnull().sum()
```

	0
type	0
director	0
cast	0
country	0
date_added	0
release_year	0
rating	4
duration	3
listed_in	0
description	0

dtype: int64

```
df['rating'] = df['rating'].fillna('Unknown')
df['duration'] = df['duration'].fillna('Unknown')
df = df.dropna(subset=['rating', 'duration'])
df.isnull().sum()
```

	0
type	0
director	0
cast	0
country	0
date_added	0
release_year	0
rating	0
duration	0
listed_in	0
description	0

dtype: int64

```
import matplotlib.pyplot as plt
import seaborn as sns

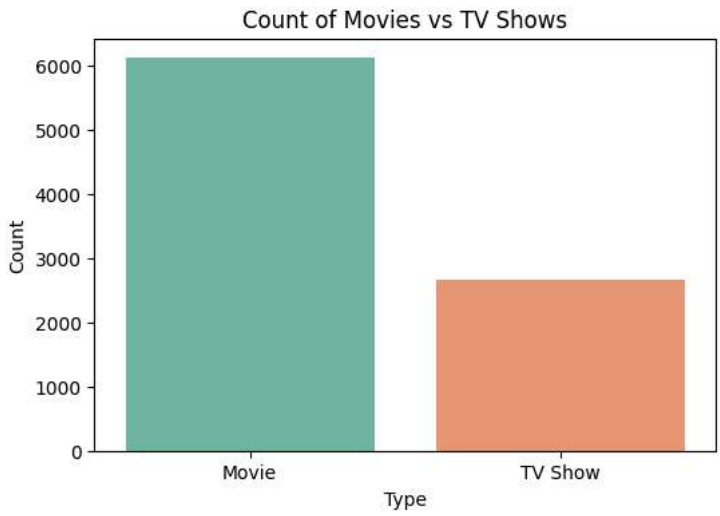
# Show plots inside notebook
%matplotlib inline
# Count of content types
plt.figure(figsize=(6,4))
sns.countplot(data=df, x='type', palette='Set2')
plt.title('Count of Movies vs TV Shows')
plt.xlabel('Type')
plt.ylabel('Count')
plt.show()

```

```
/tmp/ipython-input-20-1431812335.py:8: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set

  sns.countplot(data=df, x='type', palette='Set2')
```




```
# Most content-producing countries
top_countries = df['country'].value_counts().head(10)

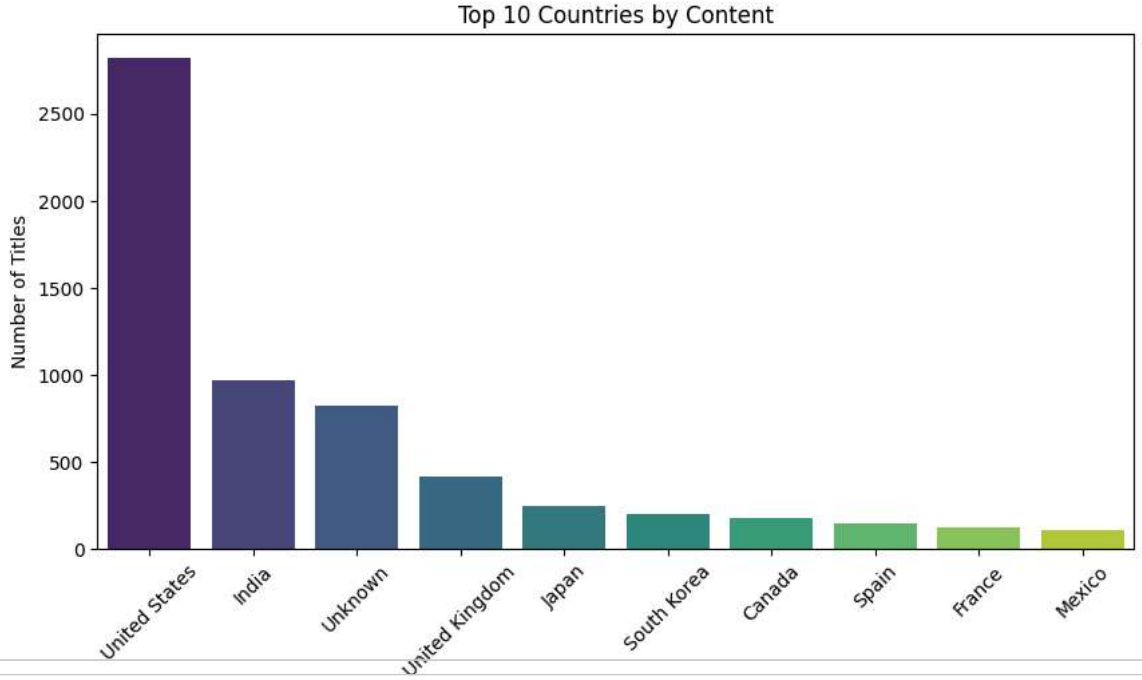
plt.figure(figsize=(10,5))
sns.barplot(x=top_countries.index, y=top_countries.values, palette='viridis')
plt.xticks(rotation=45)
plt.title('Top 10 Countries by Content')
plt.xlabel('Country')
plt.ylabel('Number of Titles')
plt.show()

```

```
/tmp/ipython-input-21-2849849447.py:5: FutureWarning:

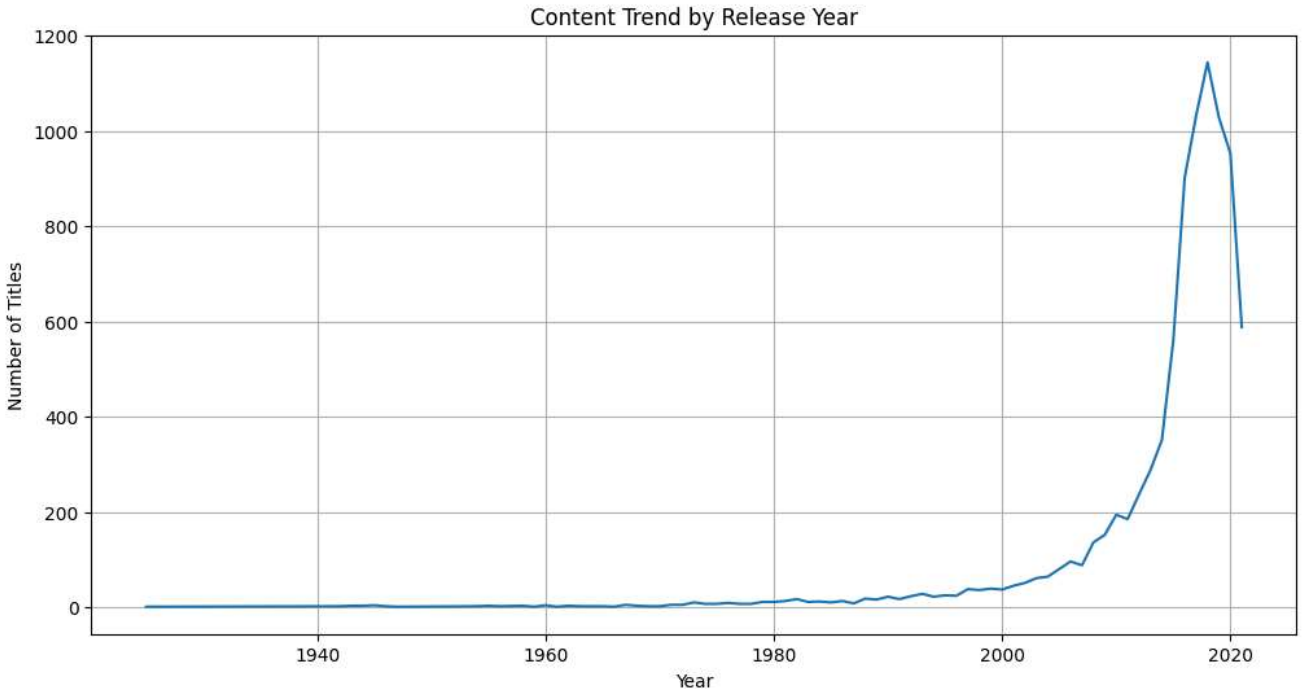
Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set

  sns.barplot(x=top_countries.index, y=top_countries.values, palette='viridis')
```




```
# Number of releases per year
plt.figure(figsize=(12,6))
df['release_year'].value_counts().sort_index().plot(kind='line')
plt.title('Content Trend by Release Year')
plt.xlabel('Year')
plt.ylabel('Number of Titles')
plt.grid(True)
plt.show()

```




```
# Most frequent ratings
plt.figure(figsize=(10,5))
sns.countplot(data=df, x='rating', order=df['rating'].value_counts().index[:10], palette='cool')
plt.title('Top Ratings')
```

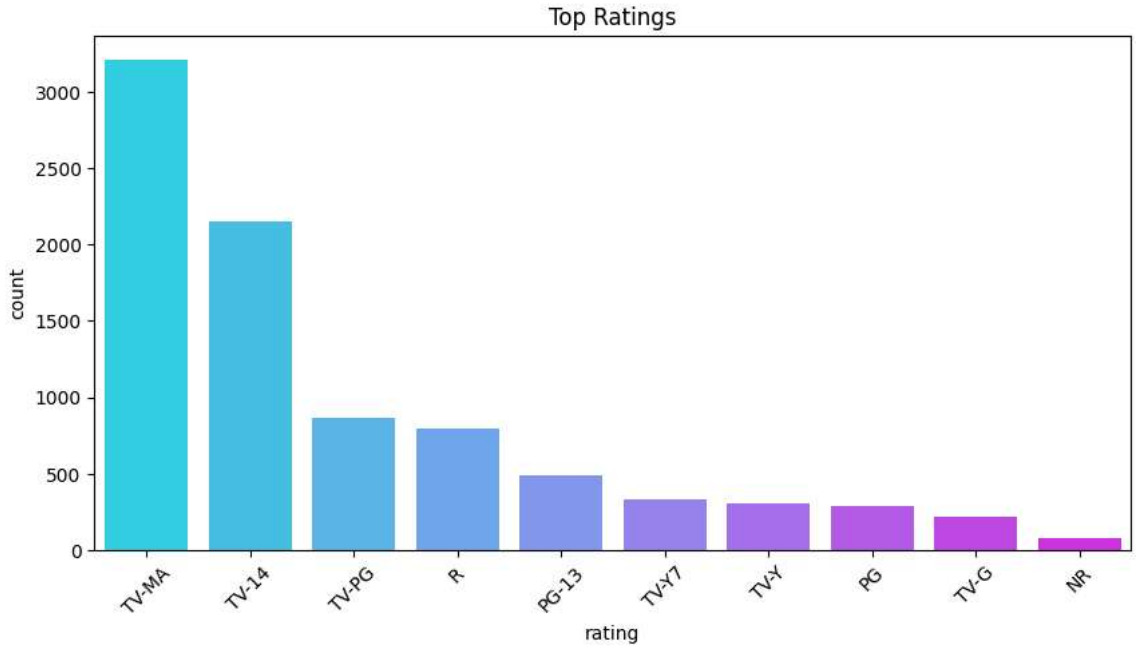
```
plt.xticks(rotation=45)
plt.show()

```

```
/tmp/ipython-input-23-3576638010.py:3: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set

  sns.countplot(data=df, x='rating', order=df['rating'].value_counts().index[:10], palette='cool')
```



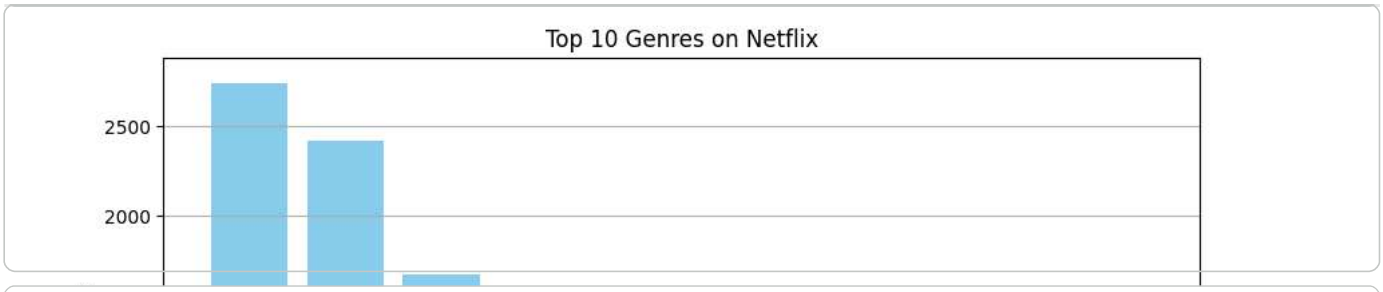

```
from collections import Counter

# Split the genres and count them
genre_list = df['listed_in'].dropna().str.split(', ')
genres = Counter([genre for sublist in genre_list for genre in sublist])

# Top 10 genres
top_genres = dict(genres.most_common(10))

# Plotting
plt.figure(figsize=(10,5))
plt.bar(top_genres.keys(), top_genres.values(), color='skyblue')
plt.title("Top 10 Genres on Netflix")
plt.xlabel("Genre")
plt.ylabel("Count")
plt.xticks(rotation=45)
plt.grid(axis='y')
plt.show()

```




```
actor_list = df['cast'].dropna().str.split(', ')
all_actors = Counter([actor for sublist in actor_list for actor in sublist])
top_actors = dict(all_actors.most_common(10))

plt.figure(figsize=(10,5))
plt.bar(top_actors.keys(), top_actors.values(), color='orange')
plt.title("Top 10 Most Frequent Cast Members")
plt.xlabel("Actor")
plt.ylabel("Appearances")
plt.xticks(rotation=45)
plt.grid(axis='y')
plt.show()

```

